

MIR MEDICAL INTELLIGENCE

Laboratory-First, Adaptive and Extensible Clinical Intelligence Research Program

Research Proposal, Scientific Methodology, Data Acquisition, ML/DL, Adaptive Test Selection, Federated Learning, Agentic AI, and Future Multimodal Expansion

Prepared: 5 September 2026 | Version 2.0 - Adaptive/Extensible Research Expansion

Primary implementation language: Python | Secondary analytical language: R

Scope: initial laboratory/non-imaging research, with an architecture deliberately designed to expand to specialized laboratory tests, longitudinal EHR data, genomics/omics, and future CT/X-ray/MRI/pathology modalities.

Core proposition: the system should reason over whatever clinically relevant evidence is available, produce its best-supported current assessment, explicitly quantify uncertainty and data coverage, identify information gaps, and request consideration of additional evidence when that evidence could materially improve the assessment. It must never fabricate unavailable observations or present a predictive signal as a definitive diagnosis.

Important safety and research position

This document proposes a research program, not a medical device specification for immediate clinical use. Any model that reaches clinical use would require appropriate clinical evidence, quality management, risk management, regulatory assessment, cybersecurity, human factors evaluation, and clinician oversight. In the European Union, software intended to provide information used for diagnostic or therapeutic decisions may fall under MDR Rule 11, and classification depends on intended use and the consequences of decisions supported by the software. [16][17]

1. Executive Summary

MIR Medical Intelligence (MIR-MI) is proposed as a research platform for laboratory-first clinical intelligence. The program will study whether routine, non-imaging data can support three related tasks: (1) detection of clinically meaningful physiological anomalies; (2) characterization of longitudinal trajectories and early deterioration; and (3) prediction of future disease outcomes, including selected cardiovascular, metabolic, renal, hepatic, hematologic, inflammatory and oncologic outcomes.

The research intentionally avoids using CT scans and X-rays as model inputs in the first program phase. This constraint creates a clear scientific question: how much clinically useful information is contained in ordinary laboratory data alone, and how much additional signal is obtained by modeling change over time rather than isolated values?

The program should be built as a sequence of increasingly difficult research tracks. Track A establishes data quality and physiological anomaly detection. Track B develops disease-specific prediction models. Track C introduces temporal and survival models. Track D studies external validity and dataset shift. Track E develops federated learning so institutions can collaborate without centralizing patient-level data. Track F introduces tightly constrained agentic AI as an orchestration and evidence-retrieval layer, never as an unconstrained autonomous diagnostician. Track G addresses prospective silent validation and clinical workflow evaluation.

The initial data strategy is deliberately heterogeneous. NHANES is best suited to population-level laboratory distributions and reference variation; MIMIC-IV and eICU provide longitudinal hospital/critical-care EHR data and outcomes; All of Us offers broad EHR and measurement data through its secure Researcher Workbench; UK Biobank is an important long-term option but currently reports that new applications are paused while platform changes are made, with new applications expected in late 2026; N3C is a further advanced option with controlled access requirements. [1][2][3][4][5][6]

The project will use reproducible cohort definitions, patient-level and temporal splits, external validation, calibration, subgroup analysis, uncertainty reporting, leakage controls, experiment tracking, model cards, and research logs. Reporting will be structured around TRIPOD+AI for prediction models, PROBAST+AI for risk of bias/applicability, and STARD/STARD-AI principles when a study is framed as diagnostic accuracy rather than future-risk prediction. [11][12][13]

Program objective

Develop and rigorously evaluate a laboratory-first clinical intelligence framework that can surface abnormal physiological patterns and calibrated future-risk signals from non-imaging clinical data, with every output traceable to data provenance, model version, evidence, uncertainty, and human review.

Primary deliverables

- A reproducible Python research repository with data ingestion, harmonization, feature engineering, modeling, evaluation, and report generation.
- A laboratory anomaly engine covering statistical, classical ML, and deep anomaly-detection approaches.
- A longitudinal trajectory engine that models level, slope, variability, persistence, and change points.
- A disease-risk benchmark suite spanning logistic regression, tree ensembles, neural networks, survival models, and calibrated ensembles.
- A federated-learning research environment using multiple simulated/partner nodes, followed by real multi-site studies when governance permits.
- A constrained clinical agent layer for evidence retrieval, tool orchestration, report drafting, provenance checking, and workflow support.

- A clinician-facing decision-support interface with explainability, uncertainty, data-coverage indicators, and auditability.
- A complete research record suitable for eventual manuscripts, theses, conference submissions, model cards, and regulatory evidence packages.

2. Research Framing and Scientific Premise

2.1 Clinical problem

Routine laboratory testing is abundant in modern healthcare, but laboratory values are commonly interpreted one result at a time and inside disease-specific workflows. The proposed research asks whether multivariable and longitudinal patterns can reveal early physiological deviation that is not obvious from a single result. The goal is not to redefine what a CBC or chemistry panel clinically means; it is to determine whether machine learning can extract additional, reproducible, and clinically useful signal from combinations and trajectories of routine measurements.

2.2 Central hypothesis

H1: longitudinal combinations of routine non-imaging laboratory measurements can identify physiologically abnormal states and estimate risk of selected future clinical outcomes better than single-test or simple threshold-based approaches, provided the research controls for confounding, missingness, dataset shift, label leakage, and calibration.

2.3 Counter-hypothesis

H0: after accounting for age, sex, care setting, missingness, and known clinical variables, laboratory-only models provide little additional predictive information for certain outcomes, or their performance is too poorly calibrated and non-generalizable to support meaningful decision support. This negative result would still be scientifically valuable and should be reported.

2.4 Clinical boundaries

- The system should not claim that a laboratory pattern proves a diagnosis.
- Cancer modules should be framed as risk/signal models, not universal cancer detectors.
- Recommendations should be phrased as clinician considerations, not autonomous treatment orders.
- The system must disclose when evidence is insufficient, data coverage is poor, or the model is outside its validated population.
- Imaging and pathology may later be used as reference standards or outcomes, but they are excluded as model inputs in the laboratory-first track.

3. Clinical Questions and Research Questions

3.1 Clinical questions

1. Can routine laboratory patterns identify physiological abnormalities beyond simple reference-range flagging?
2. Can persistent or progressive changes in laboratory values reveal early deterioration before a formal diagnosis is recorded?
3. Which laboratory combinations are associated with inflammatory, metabolic, renal, hepatic, hematologic, cardiovascular, infectious, endocrine, or nutritional states?

4. Can longitudinal lab trajectories distinguish transient abnormalities from persistent patterns?
5. Can routine laboratory data identify elevated future risk for selected diseases without using CT/X-ray images as predictors?
6. For selected cancers, can non-imaging laboratory data provide a measurable early risk signal within defined time windows?
7. How much does performance improve when historical trajectories are added to current values?
8. Can calibrated ML outputs be translated into clinician-reviewable evidence rather than opaque risk scores?

3.2 Primary methodological questions

1. Do multivariable laboratory models outperform reference-range rules and simple statistical baselines?
2. Do temporal features improve discrimination and calibration compared with cross-sectional features?
3. Which model family is most robust under missingness, class imbalance, and distribution shift?
4. Does an ensemble materially improve performance over the best single model after proper calibration?
5. Which anomaly-detection method is most stable across datasets?
6. How much performance degrades under external validation at another institution or care setting?
7. Can federated learning preserve or improve external validity while keeping raw patient-level data local?
8. Can agentic orchestration improve clinician workflow without introducing unsafe autonomous decision-making?
9. Which explanation methods are stable enough to be useful to clinicians?
10. What evidence threshold is required before a prediction is exposed in a clinical interface?

3.3 Secondary questions

Secondary analyses should examine subgroup performance, calibration drift, data missingness as signal, laboratory reference-range heterogeneity, site-specific measurement practices, model compression, uncertainty estimation, and the trade-off between sensitivity and false alerts.

4. Research Tracks

Track	Purpose	Primary data	Methods
A. Data & Normality	Build canonical laboratory data and estimate physiological variation.	NHANES + synthetic data	Schema, harmonization, missingness, reference ranges
B. Unsupervised Anomaly	Detect unusual multivariable and longitudinal profiles without disease labels.	NHANES + MIMIC-IV	Isolation Forest, LOF, One-Class SVM, AE/VAE
C. Disease Prediction	Predict selected well-defined outcomes.	MIMIC-IV / eICU / All of Us	LR, RF, XGBoost, LightGBM, CatBoost, MLP
D. Temporal Risk	Model trajectories and time-to-event outcomes.	MIMIC-IV / All of Us / UKB later	GRU, LSTM, Transformer, Cox, RSF, DeepSurv
E. Oncology	Test selected cancer-risk hypotheses without imaging inputs.	All of Us / UKB / N3C where eligible	Survival, gradient boosting, temporal DL
F. External Validation	Measure transportability across datasets and sites.	eICU, All of Us, partner data	Temporal + geographic/site validation
G. Federated Learning	Train without moving raw patient data.	Synthetic nodes first; partner sites later	FedAvg, FedProx, SCAFFOLD, personalization
H. Agentic AI	Orchestrate evidence/reasoning tools under strict controls.	All validated models	Tool-use agents, RAG, verifier agents
I. Silent Prospective	Run predictions without	Clinical partner	Prospective observational /

Track	Purpose	Primary data	Methods
Evaluation	influencing care.		silent study
J. Translation	Evaluate human-AI workflow and clinical utility.	Clinical partner	Human factors, decision curves, usability, safety

5. Dataset Strategy and Acquisition Roadmap

5.1 Dataset portfolio

Dataset	What it provides	Access route	Primary role	Ref.
NHANES	Public CDC survey/laboratory data; excellent for population variation and lab distributions.	Free/public download; follow CDC data-use and survey-weight guidance.	Baseline laboratory profiling, anomaly detection, population reference analyses.	[3][7]
MIMIC-IV	Credentialed de-identified EHR/hospital data with labs and clinical outcomes.	PhysioNet credentialing + required CITI training + DUA.	Primary clinical-development dataset; temporal trajectories and outcomes.	[1]
eICU-CRD	Multicenter critical-care database.	Credentialed access + human research training + DUA.	External validation and transportability.	[2]
All of Us	Large research program with EHR, measurements and other participant data; OMOP-standardized EHR.	Researcher Workbench registration; institutional DURA; tier-specific access.	Large-scale validation, broader population, oncology/cardiometabolic research.	[4][5][6]
UK Biobank	Large longitudinal resource with extensive phenotype, biomarkers and outcomes.	Application + scientific review + institutional process + fee/MTA after approval. New applications currently paused; late-2026 reopening expected.	Advanced validation and long-term risk studies.	[8][9]
N3C	Large U.S. EHR research enclave with multiple access levels.	N3C registration, enclave, DUA, training, data-use request; non-U.S. researchers limited to specified tiers.	Advanced external validation and federated-methodology research where eligible.	[10]
Synthea / SyntheticMass	Realistic synthetic EHR generator and synthetic population.	Open/free synthetic data and local generation.	Software development, schema tests, FL prototypes, agent evaluation.	[14]
SEER	Population cancer registry incidence, tumor characteristics and survival; not a direct laboratory dataset.	Data-access route depends on product.	Cancer outcome definitions/context; use only where linkage or compatible cohort permits.	[15]

5.2 Recommended acquisition order

1. Start with Synthea for application schema, import/export, FHIR handling, feature engineering tests, and federated-learning simulation. It is synthetic and should never be treated as a substitute for clinical evidence. [14]
2. Acquire NHANES public files and build the laboratory canonicalization layer, beginning with CBC and broad laboratory panels. The current 2021-2023 laboratory repository includes CBC and numerous additional laboratory files. [3][7]

3. Complete PhysioNet credentialing, CITI Data/Specimens Only Research training, and the MIMIC-IV data-use agreement. Do not copy or redistribute restricted patient-level files in your public repository. [1]
4. Add eICU as a deliberately external dataset rather than mixing it into the training set. Use it to test transportability. [2]
5. Apply to All of Us when institutional access is available. The Researcher Workbench is a secure cloud environment with JupyterLab and RStudio options and OMOP-standardized EHR data. [4][5][6]
6. Prepare a UK Biobank application package for the late-2026 reopening window if the intended research requires its biomarkers and outcomes. [8][9]
7. Consider N3C only when the research question and institutional eligibility justify its governance overhead. [10]

5.3 Acquisition checklist

- Create a data-source register: owner, dataset/version, access date, license/DUA, intended use, restrictions, citation, and retention period.
- Record exactly which release/version was used for every experiment.
- Keep raw restricted data outside Git and public artifact stores.
- Store only derived non-identifying aggregates or synthetic examples in the public repository.
- Document institutional approvals, IRB/ethics determinations where required, data-processing roles, and security controls before adding real institutional data.
- Create a data dictionary before model training, including source field, unit, vocabulary, reference range, missingness meaning, and transformation history.

6. Data Standardization and Clinical Data Model

6.1 Canonical internal schema

The internal model should preserve original observations and create separate normalized features. LOINC is appropriate for identifying laboratory and clinical observations; its current guidance explicitly emphasizes universal identifiers for pooling results across systems. OMOP CDM should be used as the conceptual target for broader interoperability and multi-site observational research. [20][21]

Field group	Examples
Identity	patient_id, encounter_id, observation_id
Clinical concept	LOINC code, local code, OMOP concept_id
Measurement	value, unit, UCUM unit, specimen
Reference context	reference_low, reference_high, reference_text
Time	collection_time, result_time, encounter_start, encounter_end
Source	dataset, site, laboratory source, instrument if available
Quality	missing_reason, plausibility_flag, duplicate_flag, transformation_version
Outcome linkage	outcome_definition_id, index_date, event_date, censor_date

6.2 Essential laboratory domains

- Hematology: CBC and differential; reticulocytes where available.
- Inflammation/immune: CRP, ESR, ferritin and other available markers.
- Renal/electrolytes: creatinine, eGFR-related variables, BUN/urea, sodium, potassium, bicarbonate, calcium, phosphate.
- Hepatic: ALT, AST, ALP, GGT, bilirubin, albumin, total protein.

- Metabolic/cardiovascular: glucose, HbA1c, total cholesterol, LDL, HDL, triglycerides.
- Coagulation and thrombotic markers: PT/INR, aPTT, D-dimer where present.
- Nutritional/endocrine: iron, transferrin, B12, folate, vitamin D, TSH and other available markers.
- Contextual data: age, sex, BMI, blood pressure, smoking and clinical history only when the dataset and study protocol allow.

6.3 Reference ranges and units

Never reduce all laboratory results to a single hard-coded global range. Preserve the reported reference interval when available. Normalize units using a controlled unit system such as UCUM and maintain the original value/unit alongside the normalized value. Reference ranges can vary by laboratory, method, age, sex, and clinical context.

7. Research Methodologies

7.1 Study design progression

Method	Use	Key safeguard	Primary reporting
1. Exploratory/Descriptive	Characterize distributions, missingness, unit systems, reference ranges, lab ordering patterns.	No causal claims.	Data-quality report.
2. Retrospective Cohort	Define index date, predictors available before index, and outcomes after index.	Primary design for prediction.	TRIPOD+AI + PROBAST+AI.
3. Diagnostic Accuracy	Compare an AI index test against a reference standard within a defined window.	Use only when the question is diagnostic classification.	STARD/STARD-AI principles.
4. Temporal Validation	Train on earlier time, test on later time.	Tests temporal drift.	Pre-specified cutoff.
5. External Validation	Train on one dataset/site, test on another.	Tests transportability.	No test-set tuning.
6. Prospective Silent Study	Predictions generated prospectively but hidden from clinicians.	Measures real-world distribution and drift without influencing care.	Prospective observational protocol.
7. Human-AI Study	Compare clinician workflow with and without model support.	Measures usefulness, calibration of trust, false-alert burden.	Human factors / clinical study.
8. Federated Evaluation	Run the same protocol across local nodes and aggregate permissible statistics/model updates.	Tests privacy-preserving collaboration.	Federated protocol + site harmonization.

7.2 Cohort construction

1. Define the target population and inclusion/exclusion criteria before model development.
2. Define a reproducible index date or prediction window.
3. Specify the data availability window before index date, e.g., current values only, 30/90/365-day history, or all prior measurements.
4. Define outcome events and follow-up/censoring rules before examining model performance.
5. Create patient-level splits so the same patient cannot appear across training and test sets.
6. Use temporal separation where feasible; preserve a fully untouched final test set.
7. Freeze the model and preprocessing pipeline before external validation.

7.3 Leakage prevention

Label leakage is a central risk. The study must ensure that predictors are restricted to information available at or before the prediction time. This includes avoiding post-index tests, diagnoses, procedures, or measurements that were triggered by an already-suspected disease unless the study explicitly investigates that clinical workflow. Leakage checks should be automated and reviewed by a clinician/data steward before each major experiment.

7.4 Missingness methodology

Missingness should be analyzed as data-generating behavior, not only as a preprocessing inconvenience. Compare transparent imputation approaches with models that natively handle missing values, and separately evaluate whether missingness patterns are site-specific, outcome-related, or care-intensity-related. Report the fraction of missingness for every major feature and cohort.

8. Machine Learning, Deep Learning and Statistical Model Program

8.1 Baseline ladder

Every disease-specific experiment should use a pre-specified model ladder so that deep learning is judged against strong conventional baselines rather than assumed to be superior.

Family	Models	Role
Clinical/statistical	Reference ranges, rule-based scores, linear/logistic regression	Essential baseline and interpretability.
Regularized regression	L1/L2/Elastic Net logistic regression	High-dimensional sparse/robust baseline.
Tree ensemble	Random Forest	Nonlinear interactions; robust benchmark.
Gradient boosting	XGBoost, LightGBM, CatBoost	Primary high-performance tabular candidates.
SVM	Linear/RBF SVM	Useful when sample size and feature geometry justify it.
Shallow neural	MLP	Neural tabular baseline.
Temporal neural	GRU/LSTM	Sequential laboratory trajectories.
Transformer	Time-aware Transformer / event transformer	Advanced irregular longitudinal representation.
Survival	Cox / penalized Cox	Interpretable time-to-event baseline.
Survival ML	Random Survival Forest / gradient boosting survival	Nonlinear time-to-event modeling.
Deep survival	DeepSurv or related models	Advanced survival research track.
Anomaly	Isolation Forest, LOF, One-Class SVM	Label-light anomaly detection.
Deep anomaly	Autoencoder, VAE	Latent physiological representation/anomaly score.
Ensemble	Soft voting, weighted voting, stacking	Only after single models are benchmarked and calibrated.
Uncertainty	Conformal prediction, deep ensembles, bootstrap	Decision-support uncertainty and abstention.

8.2 Recommended development sequence

1. Rule-based reference baseline.
2. Logistic regression / regularized regression.
3. XGBoost / LightGBM / CatBoost.
4. Random Forest as a diverse benchmark.
5. Calibrated ensemble.
6. MLP.
7. Temporal GRU/LSTM.

8. Time-aware Transformer.
9. Survival models.
10. Anomaly autoencoder/VAE.
11. Conformal/uncertainty layer.
12. Only then investigate multi-task or foundation-style representations.

8.3 Multi-task learning

A later research track can predict multiple related outcomes from a shared representation, e.g., cardiovascular, renal, metabolic, hepatic, and hematologic endpoints. Multi-task learning should be evaluated against independent task-specific models. Shared representations can be useful but may also cause negative transfer, so task interactions must be measured rather than assumed.

9. Model Evaluation and Statistical Analysis Plan

9.1 Primary performance metrics

- Discrimination: AUROC and area under the precision-recall curve (AUPRC).
- Classification operating points: sensitivity, specificity, PPV, NPV, F1; report confidence intervals.
- Calibration: calibration intercept/slope, calibration plots, and Brier score.
- Clinical utility: decision-curve analysis/net benefit where appropriate.
- Time-to-event: concordance index, time-dependent AUC, calibration at clinically relevant time horizons, and survival curves as appropriate.
- Anomaly detection: precision/recall against reference labels where available; stability across resamples and datasets; false-alert rate per patient or per 1,000 assessments.

9.2 Uncertainty and confidence intervals

Performance should be reported with uncertainty, preferably through bootstrap confidence intervals or another pre-specified approach. Do not rely on a single point estimate. For external validation, report the full confidence interval and calibration behavior, not just AUROC.

9.3 Subgroup evaluation

- Age bands.
- Sex.
- Care setting.
- Dataset/site.
- Baseline disease burden.
- Missingness strata.
- Other ethically and legally appropriate protected or clinically relevant groups available in the source data.

9.4 Model comparison

Use paired patient-level comparisons where suitable, such as bootstrap differences in AUROC/AUPRC, calibration comparisons, and decision-curve comparisons. Do not select a model solely because it has the highest AUROC; calibration, stability, interpretability, computational cost, external validity, and false-alert burden must be considered.

10. Explainability, Uncertainty and Clinical Communication

10.1 Explanation layers

- Data explanation: what data were available and what was missing?
- Trend explanation: which measurements changed, how quickly, and how persistently?
- Model explanation: which features most influenced this particular output?
- Population explanation: how common is this pattern in the training/validation population?
- Uncertainty explanation: how certain is the model and is it inside its validated domain?
- Action explanation: what should a clinician consider reviewing next, and what evidence supports that consideration?

10.2 SHAP and alternatives

SHAP can be used for tree and other compatible models, but explanation stability should itself be evaluated. Do not treat feature attribution as proof of causality. A feature may be predictive because it captures care intensity, disease severity, or other correlated processes. Use permutation importance, counterfactual or sensitivity analyses where appropriate to triangulate explanations.

10.3 Clinical language policy

- Preferred: “elevated risk signal”, “pattern warrants review”, “model identified persistent change”.
- Avoid: “the patient definitely has cancer” or “start treatment now” from laboratory-only AI output.
- Every high-severity alert should include an uncertainty statement and a visible “AI decision support - not a diagnosis” indicator during research use.

11. Federated Learning Research Track

11.1 Why federated learning fits this program

Healthcare data are fragmented across hospitals and health systems. Federated learning (FL) can allow multiple sites to train a shared model without centralizing raw patient records. Recent reviews show rapid growth of healthcare FL, but also emphasize that many studies remain proof-of-concept and that generalization, privacy, communication cost, and methodological quality remain important barriers. [22][23][24]

11.2 Phase-in strategy

1. Synthetic federation: create 3-10 Synthea nodes with controlled differences in age, disease prevalence, missingness, lab units and measurement noise.
2. Benchmark centralized vs federated training under IID data.
3. Introduce non-IID site differences (label prevalence, feature distributions, missingness, lab ordering patterns).
4. Compare FedAvg, FedProx and SCAFFOLD; study personalization when a global model is not optimal.
5. Add secure aggregation and privacy-preserving mechanisms where appropriate.
6. Evaluate communication cost, convergence, fairness across sites, and model drift.
7. Only then seek a real multi-site collaboration governed by data-use agreements and institutional approvals.

11.3 Recommended tooling

Flower is a strong Python-first framework for this track and provides tutorials for building federated applications and customizing strategies. NVIDIA FLARE is another option for more enterprise-oriented federated learning environments. [25]

11.4 Federated evaluation metrics

- Global model AUROC/AUPRC/calibration.
- Per-site performance and confidence intervals.
- Performance dispersion across sites.
- Worst-site and worst-subgroup performance.
- Communication rounds and bytes transferred.
- Model poisoning/robustness tests when appropriate.
- Privacy leakage assessment and threat model documentation.

11.5 Important caution

Federated learning is not automatically private. Gradients or model updates can leak information under some threat models. Therefore the research should define an attacker model and consider secure aggregation, differential privacy, access controls, audit logging, and other safeguards. FL should be treated as a distributed learning architecture, not as a blanket privacy guarantee.

12. Agentic AI Research Track

12.1 Where agents should and should not be used

Agentic AI is appropriate as an orchestration layer around deterministic tools and validated models, for example to retrieve evidence, check data quality, call a risk model, compare prior laboratory reports, generate a draft report, or route a case to a defined workflow. It should not be given unconstrained authority to diagnose, prescribe, change patient records, or bypass model thresholds.

Recent 2026 reviews describe healthcare agent research as promising but immature: a scoping review identified only a small number of rigorously evaluated studies and emphasized the limited clinical validation to date; a broader 2026 survey similarly identifies reliability, privacy, governance, and operational integration as major barriers. [26][27]

12.2 Proposed agent architecture

Agent	Purpose	Control
Data Quality Agent	Checks completeness, units, dates, duplicates, implausible values.	Read-only tools; deterministic rules.
Lab Interpretation Agent	Retrieves standardized definitions and summarizes trends.	LOINC/OMOP metadata + validated calculation library.
Model Orchestrator Agent	Selects which validated model to call based on data coverage and use case.	Allow-list only; no model generation.
Evidence Agent	Retrieves approved clinical sources and documents provenance.	RAG with curated sources and citations.
Safety/Verifier Agent	Checks claims, confidence, contradictions, and forbidden language.	Rule-based + independent verifier.
Report Agent	Produces clinician-readable summary.	Template-constrained generation.
Human Gate	Final approval before a clinical action.	Clinician remains decision-maker.

12.3 Agentic research experiments

1. Can agents correctly route laboratory-only cases to the proper validated model?
2. Can agents retrieve the correct evidence with source citations?
3. Can independent verifier agents reduce unsupported clinical claims?
4. What is the hallucination rate under adversarial and ambiguous laboratory cases?
5. Can a constrained agent reduce documentation time without degrading factual accuracy?
6. How often does the agent abstain appropriately when data are insufficient?

13. End-to-End Technical Architecture

The research platform should separate raw-data ingestion, standardized clinical representation, statistical/ML models, agentic orchestration, and the clinician interface. This separation lets you evaluate the scientific model independently of the interface or LLM layer.

13.1 Proposed architecture

Layer	Function	Suggested implementation
Input layer	PDF/CSV/Excel/FHIR/EHR extracts	Document parser, structured import, optional OCR for scanned documents
Normalization	Units, reference ranges, vocabularies	UCUM/LOINC mapping, OMOP alignment, quality checks
Patient timeline	Longitudinal observations	Temporal indexing, trajectory construction
Feature layer	Current, reference-relative, temporal, derived	Feature registry + versioned transformations
AI layer	Anomaly, classification, survival, temporal DL	scikit-learn, XGBoost, LightGBM, CatBoost, PyTorch
Calibration/uncertainty	Probabilities + abstention	Calibration models, conformal methods, bootstrapping
Explainability	Why this result?	SHAP, feature/trend evidence, model card
Agent layer	Orchestration and reporting	Tool allow-list, RAG, verifier, audit logs
Application	Clinician dashboard and reports	React + TypeScript + FastAPI
Research governance	Provenance, audit, experiment registry	MLflow, DVC, Git, metadata registry

13.2 Python project structure

```

mir-medical-intelligence/
├── README.md
├── pyproject.toml
├── uv.lock / requirements.lock
├── CITATION.cff
├── LICENSE
├── CHANGELOG.md
├── docs/
│   ├── research_protocol.md
│   ├── research_questions.md
│   ├── cohort_definitions/
│   ├── data_sources.md
│   ├── data_dictionary.md
│   ├── model_cards/
│   ├── risk_management.md
│   ├── regulatory_notes.md
│   └── papers/
├── data/          # never commit restricted patient-level data
│   ├── raw/
│   ├── interim/
│   └── processed/

```

```

├── metadata/
├── notebooks/
│   ├── 01_data_audit/
│   ├── 02_cbc_baseline/
│   ├── 03_anomaly_detection/
│   ├── 04_supervised_benchmarks/
│   ├── 05_temporal_models/
│   ├── 06_survival/
│   ├── 07_external_validation/
│   ├── 08_federated_learning/
│   └── 09_agent_evaluation/
├── src/mir_medical/
│   ├── ingestion/
│   ├── schemas/
│   ├── harmonization/
│   ├── cohort/
│   ├── features/
│   ├── anomaly/
│   ├── models/
│   ├── temporal/
│   ├── survival/
│   ├── calibration/
│   ├── uncertainty/
│   ├── explainability/
│   ├── federated/
│   ├── agents/
│   ├── evaluation/
│   └── reporting/
├── configs/
│   ├── datasets/
│   ├── cohorts/
│   ├── models/
│   └── experiments/
├── pipelines/
│   ├── ingest.py
│   ├── harmonize.py
│   ├── build_cohort.py
│   ├── build_features.py
│   ├── train.py
│   ├── evaluate.py
│   └── generate_report.py
├── tests/
├── app/
│   ├── backend/
│   └── frontend/
├── models/
│   ├── registry/
│   ├── checkpoints/
│   └── archived/
├── reports/
│   ├── figures/
│   ├── tables/
│   └── experiments/

```

13.3 Recommended Python libraries

Area	Libraries
Data	pandas, polars, numpy, scipy, pyarrow
Statistics	statsmodels, scipy

Area	Libraries
Classical ML	scikit-learn
Boosting	xgboost, lightgbm, catboost
Deep learning	torch, pytorch-lightning or equivalent
Time series/survival	sktime where appropriate, lifelines, scikit-survival
Explainability	shap
Experiment tracking	mlflow
Data/versioning	dvc or lakeFS, Git
Federated	Flower; NVIDIA FLARE where required
FHIR/clinical data	FHIR client libraries, terminology services; validate against HL7/LOINC/OMOP specifications
API	FastAPI, pydantic
UI	React + TypeScript + Plotly
Testing/quality	pytest, ruff, mypy, pre-commit

13.4 Optional R environment

R should be retained as a secondary statistical environment for biostatistics, survival analysis, epidemiology, sensitivity analyses, and publication-quality statistical graphics. Recommended packages include tidyverse, data.table, survival, tidymodels, rms, caret or modern equivalents, ggplot2, and survey for complex NHANES analyses. The canonical production/research pipeline remains Python to minimize duplication.

14. Core Experimental Program

ID	Experiment	Question	Data	Methods	Output
EXP-001	CBC data audit	What are the distributions, missingness and reference-range patterns?	NHANES	Descriptive stats, QC	Data dictionary + baseline report
EXP-002	Population anomaly detection	Can unsupervised models detect unusual lab profiles?	NHANES	z-score, Mahalanobis, IF, AE	Anomaly benchmark
EXP-003	Clinical abnormality prediction	Can models predict a defined outcome from routine labs?	MIMIC-IV	LR, RF, XGB, LGBM, CatBoost	Benchmark + calibration
EXP-004	Trajectory advantage	Do longitudinal features outperform current values?	MIMIC-IV	Static vs temporal features	Paired model comparison
EXP-005	External validation	Does the locked model transport?	eICU	Frozen model	External validity report
EXP-006	Survival prediction	Can labs estimate 1/3/5-year risk for selected outcome?	All of Us/UKB/N3C as eligible	Cox, RSF, DeepSurv	Time-to-event analysis
EXP-007	Cancer signal	Do labs provide an early risk signal for selected cancer?	All of Us/UKB/N3C as eligible	Temporal + survival ML	Oncology feasibility paper
EXP-008	Federated benchmark	Does FL match centralized training?	Synthetic nodes	FedAvg/FedProx/SCAFFOLD	FL methods paper
EXP-009	Agent routing	Can constrained agents safely orchestrate models?	Synthetic + retrospective cases	Tool-use + verifier	Agent safety report
EXP-010	Silent prospective study	What happens in live workflow without influencing care?	Clinical partner	Prospective silent validation	Deployment readiness report

15. First 90 Days - Concrete Execution Plan

Days 1-15: scientific and data foundation

- Freeze research charter, scope, hypotheses, outcomes, and exclusions.

- Create Git repository and research log template.
- Install Python environment; create project skeleton and test suite.
- Download synthetic Synthea data and build FHIR/CSV import pipeline.
- Download NHANES public CBC and selected laboratory files.
- Write the first data dictionary and unit-normalization registry.

Days 16-30: laboratory representation

- Build canonical laboratory table.
- Implement quality flags and unit normalization.
- Implement reference-range-relative features.
- Create missingness report.
- Create first patient/timepoint and trajectory data structures.
- Document every transformation in code and metadata.

Days 31-60: anomaly research

- Implement robust z-score and Mahalanobis baselines.
- Implement Isolation Forest, LOF and One-Class SVM.
- Implement an autoencoder baseline.
- Compare stability across resampling and population strata.
- Create first anomaly report and visualization layer.

Days 61-90: MIMIC + supervised baseline

- Complete PhysioNet credentialing/CITI/DUA steps and access MIMIC-IV. [1]
- Create an initial cohort for one well-defined clinical outcome.
- Train logistic regression, random forest, XGBoost, LightGBM and CatBoost.
- Set patient-level and temporal validation splits.
- Generate AUROC, AUPRC, calibration, confidence intervals, SHAP and error analysis.
- Write Research Report v0.1 documenting the full experiment.

16. 24-Month Research Roadmap

Period	Focus	Milestone
Months 1-3	Infrastructure + NHANES + Synthea	Canonical schema, data dictionary, anomaly baseline
Months 4-6	MIMIC-IV baseline prediction	First disease-specific benchmark + calibration
Months 7-9	Temporal modeling	Trajectories, GRU/LSTM, survival baseline
Months 10-12	External validation	eICU validation + dataset shift paper
Months 13-15	All of Us / advanced outcomes	Broader population + selected disease/cancer outcomes
Months 16-18	Federated learning	Synthetic non-IID federation + privacy threat model
Months 19-21	Agentic orchestration	Tool-use, RAG, verifier, safety benchmark
Months 22-24	Clinical workflow readiness	Silent prospective design, model cards, risk management, manuscript package

17. Research Paper and Evidence Package Strategy

17.1 Suggested manuscripts

Paper	Topic	Design	Data	Primary contribution
Paper 1	Laboratory-only physiological anomaly detection	Cross-sectional + anomaly methods	NHANES	Descriptive/anomaly validity
Paper 2	Longitudinal laboratory trajectories	Retrospective cohort	MIMIC-IV	Trajectory prediction
Paper 3	External validation and dataset shift	External validation	MIMIC-IV -> eICU	Transportability
Paper 4	Laboratory-only cancer risk signals	Retrospective cohort / survival	All of Us or UKB/N3C if eligible	Oncology feasibility
Paper 5	Federated laboratory risk modeling	Methodological / multi-site simulation	Synthea + partner nodes	Privacy-preserving ML
Paper 6	Constrained agentic clinical intelligence	Benchmark + safety evaluation	Synthetic + retrospective cases	Agent reliability and governance

17.2 Reporting frameworks

Use TRIPOD+AI for prediction model development/evaluation, PROBAST+AI for bias and applicability assessment, and STARD/STARD-AI principles when an experiment evaluates diagnostic accuracy against a reference standard. These frameworks emphasize transparent cohort definition, predictor/outcome handling, preprocessing, model development, validation, fairness, explainability and appropriate interpretation. [11][12][13]

17.3 Research package for every experiment

- Research question and pre-specified hypothesis.
- Data-source/version record.
- Cohort SQL or reproducible cohort code.
- Feature specification and data dictionary version.
- Training/validation/test split definitions.
- Model configuration and random seed.
- Evaluation script and raw metrics output.
- Confidence intervals and calibration plots.
- Subgroup and error analysis.
- Model card and limitations.
- Researcher interpretation and next experiment.

18. Ethics, Privacy, Safety and Regulatory Roadmap

18.1 Research governance

Health data are a special category of personal data under EU data protection law, and scientific research involving such data requires an appropriate lawful basis and safeguards. The EDPB published Guidelines 1/2026 on processing personal data for scientific research, including safeguards for large-scale processing of special-category data and attention to national-law conditions. [18][19]

18.2 EU medical-device implications

The intended use must be documented before software is positioned as a clinical product. Under MDR Rule 11, software intended to provide information used to take diagnostic or therapeutic decisions is generally class IIa, with escalation to class IIb or class III depending on the potential consequences of those decisions. The European Commission updated MDCG 2019-11 on software qualification/classification in June 2025. [16][17]

18.3 EU AI Act status as of 5 September 2026

The EU AI Act generally applies from 2 August 2026, while AI systems embedded in regulated products under Annex I, which can include medical devices, have an extended transition for the high-risk rules to 2 August 2028 under the current text. The legal position should be rechecked at every product milestone because implementation guidance and amendments evolve. [17][18]

18.4 ISO 14971 and lifecycle risk management

ISO 14971:2019 remains current and provides a lifecycle risk-management process covering medical-device software and in-vitro diagnostic software. The research program should create a preliminary hazard log early rather than waiting for commercialization. [17]

18.5 Proposed hazard classes

Hazard	Example	Control
False negative	Model misses a serious risk signal.	Uncertainty, sensitivity-focused thresholds, abstention, clinical review.
False positive	Excessive alerts or unnecessary workup.	Alert-rate monitoring, specificity/PPV, decision curves, workflow limits.
Dataset shift	Model fails at another site.	External validation, site monitoring, recalibration protocol.
Label leakage	Performance inflated by future information.	Index-date firewall, automated temporal checks, clinician review.
Reference-range error	Wrong interpretation due to unit/range mismatch.	Preserve source range, UCUM, validation tests.
Automation bias	Clinician over-trusts AI.	Human factors testing, explanation, uncertainty, explicit AI status.
Agent hallucination	Generated narrative contains unsupported claim.	Evidence citations, verifier agent, constrained templates, human sign-off.
Privacy leakage	Model/data expose sensitive information.	Access controls, encryption, audit logs, DUA, FL/privacy methods where appropriate.

19. Clinician Interface and User Experience

19.1 Core views

- Patient overview: data availability, last laboratory assessment, key active signals.
- Laboratory timeline: trends, abnormality persistence, change points, comparison to prior measurements.
- Risk matrix: disease-system risk signals with calibrated probabilities and confidence/coverage.
- Explanation view: top contributing features, trend evidence, reference ranges, and model version.
- Clinical review: suggested considerations, evidence sources, and “why this was flagged”.
- Audit/research view: dataset, model version, feature version, prediction timestamp, and decision trace.

19.2 Data-coverage indicator

The interface should visibly distinguish well-supported assessments from low-coverage assessments. For example: “Hematology: high coverage; Metabolic: moderate; Oncology: insufficient evidence.” A system should not create the appearance of a full-body medical assessment when only one CBC is available.

19.3 Example report language

Example research-output phrasing: “The model identified a persistent change in hemoglobin and RDW over the available observation window. This pattern is non-specific and does not establish a diagnosis. The result is shown as a risk signal for clinician review. The model was trained on [dataset/version], validated on [dataset/version], and has a calibrated performance of [metric + CI] in the validated cohort.”

20. Optional R Companion Structure

```
mir-medical-r/
├── renv.lock
├── R/
│   ├── cohort_helpers.R
│   ├── survival_models.R
│   ├── survey_weights_nhanes.R
│   ├── calibration.R
│   └── publication_figures.R
├── analysis/
├── reports/
└── tests/testthat/
```

Keep one source of truth for cohort definitions and feature specifications. R should consume exported analysis-ready tables or shared code outputs rather than independently redefining the cohort.

21. Governance, Reproducibility and Model Lifecycle

21.1 Experiment registry

Each experiment receives an immutable ID, dataset version, cohort version, feature version, model version, random seed, code commit hash, configuration file, metrics, and artifacts. MLflow can track runs, parameters and artifacts; Git records code; DVC or equivalent can track data metadata and derived datasets.

21.2 Model cards

Field	Required content
Intended use	What the model is for and what it is not for.
Population	Inclusion/exclusion, demographics, care setting.
Input features	Exact measurements, windows, preprocessing.
Training data	Dataset/version and access restrictions.
Validation	Internal, temporal, external; sizes.
Performance	Discrimination, calibration, uncertainty.
Subgroups	Observed performance differences.
Failure modes	Known false-positive/negative patterns.
Limitations	Generalizability, missingness, confounding, reference standard.
Monitoring	Drift, recalibration, incident response.
Versioning	Model ID, code commit, training timestamp.

21.3 Change control

No model should silently change in production. Any new model or recalibration requires a version increment, validation report, approval record, and documented impact assessment. A model may be retired if calibration, subgroup performance, or data drift exceed pre-defined limits.

22. Research Risks and Mitigation Register

Risk	Impact	Mitigation
Insufficient signal	Low or non-specific laboratory signal for some diseases.	Focus on defined outcomes; publish negative results; use imaging/pathology only as reference standards, not predictors in the lab-first track.
Confounding	Model learns care intensity instead of biology.	Temporal index, causal sensitivity analyses, site validation, clinical review.
Spectrum bias	Training population differs from target population.	Multiple datasets and external validation.
Class imbalance	Rare outcomes distort metrics.	AUPRC, class-weighting, focal loss where justified, sampling only inside training.
Overfitting	Too many features and experiments.	Pre-specification, nested CV where needed, untouched test set, model simplicity baselines.
Dataset shift	Performance collapses across sites.	External validation and federated/site-aware methods.
Agent overreach	LLM invents unsupported medical claims.	Tool allow-list, evidence citations, verifier, hard refusal/abstention.
Regulatory mismatch	Research prototype accidentally marketed as clinical device.	Clear research labeling, intended-use control, regulatory review before deployment.
Privacy breach	Improper use or release of restricted data.	DUAs, secure storage, access control, no raw-data commits, privacy review.

23. Success Criteria

23.1 Scientific success

- At least one laboratory-only anomaly model demonstrates reproducible detection against a defensible reference definition.
- At least one longitudinal model shows whether historical trajectories add predictive value over cross-sectional features.
- At least one model is externally validated with calibrated performance and documented transportability limitations.
- At least one cancer-risk question is tested using a pre-specified outcome window and leakage-controlled cohort.
- Federated experiments quantify the trade-off between centralized and decentralized training.
- Agent experiments quantify factuality, abstention, routing accuracy, and safety rather than subjective “looks good” judgments.

23.2 Engineering success

- Every model result is reproducible from a versioned configuration.
- Dataset ingestion and feature engineering have automated tests.
- The clinical interface displays model provenance and uncertainty.
- The research platform can run on synthetic data without restricted access.
- No restricted dataset is accidentally committed to source control or shared outside its permitted environment.

24. Immediate Acquisition Checklist

Step	Action	Deliverable
1	Create research identity/account	PhysioNet, All of Us, relevant institutional accounts.
2	Complete CITI training	Required for MIMIC/eICU; retain certificates and dates.
3	Execute MIMIC/eICU DUA	Follow current PhysioNet process.
4	Download NHANES files	Start with CBC_L and related laboratory files; preserve documentation.
5	Install synthetic environment	Run Synthea locally and generate FHIR/CSV test cohorts.
6	Assess All of Us eligibility	Check whether your institution has a current DURA and which tier is available.
7	Prepare UKB concept note	Develop a concise scientific proposal for the late-2026 application window if needed.
8	Define first outcome	Choose one common, well-labeled clinical endpoint for MIMIC-IV.
9	Create protocol before modeling	Freeze index date, outcome, predictors, split strategy and primary metrics.
10	Start experiment EXP-001	Population laboratory audit + data dictionary + anomaly baseline.

25. Reference Sources and Access Links

- [1] PhysioNet. MIMIC-IV v3.1; credentialed access, CITI training, DUA. [Source](#)
- [2] PhysioNet / MIT LCP. eICU Collaborative Research Database v2.0; credentialed access, training and DUA. [Source](#)
- [3] CDC/NCHS. NHANES Continuous Laboratory Data 2021-2023. [Source](#)
- [4] NIH All of Us. Data access tiers and Researcher Workbench. [Source](#)
- [5] NIH All of Us. Researcher Workbench and current data collections. [Source](#)
- [6] NIH All of Us. Data sources and OMOP standardization. [Source](#)
- [7] CDC/NCHS. CBC_L documentation, 2021-2023. [Source](#)
- [8] UK Biobank. Apply for access; current 2026 application status. [Source](#)
- [9] UK Biobank Community. How to complete an access application. [Source](#)
- [10] NCATS NIH. N3C data overview and access requirements. [Source](#)
- [11] BMJ. TRIPOD+AI reporting guideline for prediction models using regression or ML. [Source](#)
- [12] BMJ. PROBAST+AI: quality, risk of bias and applicability. [Source](#)
- [13] BMJ Open. STARD-AI protocol and diagnostic-accuracy reporting considerations. [Source](#)
- [14] Synthea. Synthetic Patient Population Simulator and SyntheticMass. [Source](#)
- [15] NCI SEER. SEER incidence data and cancer registry resources. [Source](#)
- [16] European Commission / MDCG. MDCG 2019-11 rev.1: software qualification and classification, June 2025. [Source](#)
- [17] EUR-Lex. Regulation (EU) 2017/745, including MDR Rule 11. [Source](#)

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- [19] EDPB. Guidelines 1/2026 on processing personal data for scientific research. [Source](#)
- [20] LOINC. About LOINC and laboratory/clinical observation identifiers. [Source](#)
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- [23] PubMed. Implementing federated learning in healthcare: pitfalls and recommendations. [Source](#)
- [24] PubMed. Methodological gaps limiting federated learning in healthcare, 2026. [Source](#)
- [25] Flower. Federated learning framework and tutorials. [Source](#)
- [26] PubMed. The role of agentic artificial intelligence in healthcare: scoping review, 2026. [Source](#)
- [27] PubMed. A comprehensive survey of AI agents in healthcare, 2026. [Source](#)
- [28] WHO. Regulatory considerations on artificial intelligence for health. [Source](#)
- [29] WHO. Ethics and governance of AI for health / large multi-modal models. [Source](#)
- [30] ISO. ISO 14971:2019 medical-device risk management; confirmed current in 2025. [Source](#)

Appendix A. Example Research Log Template

Experiment ID:
 Date:
 Research question:
 Hypothesis:
 Dataset + version:
 Access/DUA reference:
 Cohort definition:
 Index date definition:
 Outcome definition:
 Predictor window:
 Excluded information / leakage controls:
 Preprocessing version:
 Feature version:
 Models:
 Hyperparameters/config:
 Split strategy:
 Primary metric:
 Secondary metrics:
 Calibration method:
 Confidence interval method:
 Subgroups:
 Results:
 Error analysis:
 Limitations:
 Interpretation:
 Decision: keep / revise / reject:
 Next experiment:
 Code commit:
 Model version:

Appendix B. Minimum Model Card Template

Model name/version: _____

Intended use: _____

Not intended for: _____

Training population: _____

Features: _____

Training dataset/version: _____

Internal validation: _____

External validation: _____

AUROC/AUPRC: _____

Calibration: _____

Known failure modes: _____

Fairness/subgroups: _____

Uncertainty/abstention: _____

Monitoring plan: _____

Clinical/regulatory status: research only / under evaluation / approved use (must be documented)

Appendix C. Research Position Statement

The central scientific opportunity is not to prove that a CBC can diagnose every disease. It is to measure, rigorously and honestly, how much information routine non-imaging laboratory data contain about physiological state and future disease risk, how much of that information persists across time and across institutions, and whether models can convert that signal into calibrated, explainable, clinician-useful decision support. A successful program may discover strong signals for some outcomes, weak signals for others, and no meaningful signal for still others. The methodology is designed to make all three outcomes useful scientific findings.

Major Program Expansion - Adaptive, Evidence-Seeking and Multimodal Design

Purpose of this expansion. The original proposal established a laboratory-first clinical intelligence research program. This version adds a central capability: the system must operate under incomplete and changing evidence, deliver a current assessment from the information actually provided, quantify what it can and cannot infer, and identify additional measurements or records that could reduce uncertainty. The research program also treats laboratory inputs as an extensible ontology rather than a fixed CBC/CMP/lipid checklist and defines a migration path toward imaging and other modalities.

26. Updated System Operating Principle

26.1 Evidence-in-hand first

Every patient encounter begins with an evidence inventory. The engine must analyze the available observations immediately rather than fail because a preferred panel is absent. A CBC alone should trigger hematological analysis, while the system clearly labels metabolic, cardiovascular or oncological conclusions as limited when required evidence is missing.

26.2 Partial observability is a design requirement

The patient state is represented as a partially observed clinical state: observed measurements, unavailable measurements, pending measurements, historical measurements, timestamps, reference intervals, provenance and data quality. Missing is never interpreted as normal, and imputed values are never displayed as observed facts.

26.3 Every inference has three components

- Current assessment: what can be inferred from the evidence currently available.
- Uncertainty statement: how strongly the system can support that inference, including data coverage and model uncertainty.
- Evidence gap: what additional measurement, history, prior report or other evidence could materially improve the assessment.

26.4 The system may be inconclusive without being silent

An indeterminate output is a valid research outcome. The engine should be allowed to abstain from a specific classification while still reporting observed abnormalities and explaining what prevents a stronger conclusion. This supports safer human-in-the-loop use and enables research on selective prediction/abstention.

27. Adaptive Laboratory and Evidence Acquisition Research Track

27.1 Central research question

Can an uncertainty-aware clinical intelligence system determine when additional laboratory or clinical information is likely to materially reduce uncertainty, identify the highest-value next evidence source, and update its assessment after new evidence arrives?

27.2 Research sub-questions

1. How does predictive performance degrade as clinically relevant measurements are withheld?
2. Which missingness-aware representations are most robust: imputation, masks, missingness indicators, native tree-model handling, or temporal models?
3. Can the model identify when available evidence is sufficient for a particular clinical task?
4. Can expected value of information (EVI) or information-gain methods prioritize additional tests?
5. Can adaptive test selection improve uncertainty and calibration without increasing unnecessary testing?
6. How does adaptive acquisition behave when laboratories have different test availability patterns?
7. Can a clinician override or reject the proposed additional test without degrading traceability?
8. Does the system become safer when it abstains rather than forcing a prediction?

27.3 Formal experimental setup

Use a dataset in which a reasonably complete set of measurements and outcomes is available. Create controlled missingness scenarios by masking one test, groups of tests, random measurements, clinically selected measurements, and site-specific patterns. Compare the following policies: (A) no adaptation; (B) conventional imputation; (C) rule-based test recommendation; (D) model-driven uncertainty; (E) EVI/information-gain acquisition. Compare discrimination, calibration, uncertainty reduction, number of requests, and resource proxies.

27.4 Recommended test-selection outputs

- High-value next evidence: the most informative missing measurement or document for the current question.
- Reason: why the additional evidence is relevant to the current uncertainty.
- Expected impact: predicted reduction in uncertainty or expected change in decision confidence.
- Current limitation: what remains unresolved without the evidence.
- Clinician control: request is advisory and does not autonomously order a test.

28. Extensible Laboratory Ontology - Do Not Limit the Model to Common Tests

The initial training program should use the major laboratory families already identified in the proposal, but the data model must not contain a closed list such as CBC/CMP/lipids only. New analytes should be onboarded through metadata-driven registration rather than code rewrites.

28.1 Example laboratory families for future onboarding

Family	Examples / onboarding scope
Hematology	CBC, differential, reticulocytes, morphology-derived measures, coagulation studies
Inflammation / immune	CRP, ESR, ferritin, fibrinogen, immunoglobulins, complement, autoantibodies where ethically and clinically appropriate
Cardiac	troponin, BNP/NT-proBNP and other validated cardiac biomarkers

Family	Examples / onboarding scope
Metabolic	glucose, HbA1c, insulin, lipids, uric acid and broader metabolic chemistry
Renal	creatinine, eGFR, BUN/urea, albuminuria and urinalysis
Hepatic	ALT, AST, ALP, GGT, bilirubin, albumin and related markers
Endocrine	TSH, free thyroid hormones, cortisol, reproductive and other endocrine markers
Nutrition	iron, ferritin, transferrin, B12, folate, vitamin D and related measures
Infectious disease	serology, antigen/antibody tests and pathogen-related measurements where datasets permit
Oncology-related	tumor-associated markers and hematology/chemistry signals, treated as non-specific risk evidence rather than stand-alone diagnosis
Specialty / rare	genetic, metabolic, rheumatologic, neurologic, toxicology, transplant and other specialized assays as future research permits

28.2 Metadata-driven onboarding

Each new analyte should be registered with: standardized concept identifier where available (for example LOINC), local code(s), display name, synonyms, unit vocabulary, permitted unit conversions, specimen type, reference-range policy, age/sex/context modifiers, laboratory provenance, known derived features, data-quality rules, and model eligibility flags. The model should therefore grow by adding validated feature definitions and retraining rather than by hard-coding a fixed panel.

29. Missingness-Aware Model Architecture

The research baseline should treat missingness as a first-class feature of the clinical environment. For each measurement, maintain both the value and an observation state. Suggested state flags are: observed, unavailable, not ordered, ordered/pending, invalid, censored/withheld for privacy, and not applicable.

29.1 Feature representation

- Value vector: normalized observed measurements.
- Observation mask: 1/0 or categorical status for availability.
- Age-of-measurement: time since the observation was collected.
- Provenance: source system/laboratory/measurement context when permitted.
- Reference context: reported low/high interval, unit and relevant demographic context.
- Temporal summary: previous value, change, slope, persistence, count of abnormal observations, and recency.

29.2 Model families to benchmark

- Native missing-value gradient boosting: XGBoost, LightGBM, CatBoost.
- Explicit imputation plus missingness indicators: median, KNN, MICE and model-based approaches.
- Mask-aware neural networks using value + mask + time-since-measurement.
- Temporal models: GRU/LSTM/Transformer with irregular-event representation.
- Probabilistic approaches: calibrated Bayesian/uncertainty methods where feasible.
- Selective prediction: confidence thresholds and abstention policies.

30. Research Methodology for Adaptive Testing

30.1 Cohort and index design

For each clinical task define an index time T_0 , a predictor observation window ending at T_0 , and a future outcome window after T_0 . No information collected after T_0 may enter the predictor features. When a test is selected as an adaptive candidate, the model must only use information available before the hypothetical acquisition decision.

30.2 Sequential decision experiment

Simulate the clinical loop: observe partial data -> predict -> estimate uncertainty -> choose next evidence candidate -> reveal that evidence from the hidden complete record -> update patient state -> repeat until a stopping condition is met. This allows the research team to evaluate whether adaptive acquisition improves usefulness with fewer additional measurements.

30.3 Stopping rules

- Sufficient confidence for the defined task.
- No additional candidate provides a pre-specified minimum expected benefit.
- Data coverage threshold achieved.
- Maximum number of additional requests reached.
- Clinician rejects further acquisition.
- Safety rule or contraindication prevents recommending a test.

31. Research Materials and Evidence Package - What Must Be Produced

A complete research program needs more than code and a model. The following artifacts should be produced and versioned as first-class research materials.

Artifact	Purpose
Research protocol	Purpose, hypotheses, endpoints, cohorts, inclusion/exclusion, predictors, outcomes, validation, stopping rules.
Literature review matrix	Question, search strategy, sources, evidence strength, gaps, contradictions, date checked.
Data access dossier	Dataset owner, version, access route, training/DUA/IRB requirements, allowed use, citation, acquisition date.
Data dictionary	All variables, units, codes, missingness, provenance and transformations.
Cohort specification	SQL/Python definitions of eligible participants and index dates.
Feature specification	Exact definitions and formulas for every predictor.
Statistical analysis plan	Primary/secondary outcomes, metrics, CI methods, subgroup analysis, sensitivity analyses.
Experiment registry	Immutable experiment ID, dataset/version, code commit, config, results, interpretation.
Model card	Intended use, population, training data, performance, calibration, limitations, risks.
Dataset card / provenance record	Source, licensing/access, cohort scope, known biases, transformations.
Risk register	Clinical, technical, privacy, security, fairness and deployment hazards.

Artifact	Purpose
Reproducibility package	Environment lockfile, seeds, configs, scripts, tests, figures and tables.
Error-analysis report	False positives, false negatives, subgroup errors, missingness failures, distribution shift.
Clinical interpretation guide	Permitted wording, uncertainty language, non-diagnostic caveats, escalation boundaries.
Human-factors study materials	Workflow scenarios, usability tasks, clinician feedback instruments.
Regulatory position memo	Intended purpose, software qualification/classification analysis, evidence plan.
Manuscript package	Structured abstract, methods, results, figures, supplementary material and reporting checklists.

32. Research Work Packages and Task Breakdown

Work package	Core tasks
WP0 - Governance	Research protocol, ethics determination, data use agreements, privacy plan, data management plan, authorship plan, reproducibility policy.
WP1 - Data acquisition	Dataset registration, access applications, downloads/exports, provenance, integrity checks, storage and versioning.
WP2 - Clinical ontology	LOINC/OMOP/ICD/SNOMED mapping where appropriate, synonyms, units, reference intervals, specialty-test onboarding.
WP3 - Data quality	Missingness, duplicates, impossible values, unit errors, timestamp problems, lab-to-lab differences.
WP4 - Anomaly science	Normal variation, anomaly detectors, persistence, trend analysis, false-alarm analysis.
WP5 - Predictive modeling	Baselines, tree ensembles, deep learning, survival models, multi-task models.
WP6 - Adaptive evidence acquisition	Uncertainty, abstention, EVI, next-test recommendations, sequential simulation.
WP7 - Explainability	Global/local importance, trend explanations, confidence, counterfactual cautions.
WP8 - External validation	Temporal/site split, cross-dataset validation, calibration, subgroup/fairness analysis.
WP9 - Federated learning	Simulated clients first; real-site collaboration later; non-IID and heterogeneous feature research.
WP10 - Agentic layer	Tool orchestration, evidence retrieval, model routing, verification and report drafting with constrained actions.
WP11 - Multimodal expansion	Add imaging/pathology when laboratory-only scientific questions have matured.
WP12 - Human factors and clinical evaluation	Clinician workflow simulation, silent prospective validation, usability and safety evaluation.
WP13 - Publication	Paper-specific analysis packages, reporting checklists, supplementary materials and release packages.

33. Dataset Acquisition Roadmap - Exact Sequence

33.1 Stage 1 - Synthetic and open data

1. Synthea: use synthetic FHIR/CSV data to build ingestion, patient-state, longitudinal, missingness and adaptive-acquisition plumbing without privacy restrictions.
2. NHANES: use publicly available laboratory files to study population laboratory variation, normal-range behavior and selected outcomes available in the survey ecosystem.
3. Deliverable: a fully reproducible pipeline that works without any credentialed dataset.

33.2 Stage 2 - Credentialed clinical data

4. MIMIC-IV: complete PhysioNet credentialing, required training and DUA; build the first serious hospital/EHR cohort and benchmark models.
5. eICU: complete the equivalent credentialed access workflow; reserve the dataset primarily for external validation and heterogeneity research.
6. Deliverable: locked development model + external validation package.

33.3 Stage 3 - Broad longitudinal EHR research

7. All of Us: determine institutional eligibility/DURA, obtain researcher authorization, define the workspace and data tier, then study broader EHR and laboratory generalization.
8. N3C and similar controlled clinical networks: pursue only when the research question justifies the governance and access overhead.
9. Deliverable: multi-population transportability study.

33.4 Stage 4 - Future imaging

10. TCIA: begin with public, de-identified DICOM collections for cancer imaging research and develop an imaging data ingestion/QA branch.
11. UK Biobank imaging: pursue as an advanced research option when applications are open and eligibility/institutional requirements are satisfied. As of 1 July 2026, UK Biobank reported that new applications were paused and intended to accept new applications in late 2026.
12. Task-specific imaging datasets such as brain-tumor MRI/CT collections can be added for segmentation/classification research, but must be treated as separate cohorts and not automatically merged with laboratory cohorts.
13. Deliverable: multimodal research branch in which imaging can be added without redesigning the laboratory engine.

Dataset	Domain	Access	When	Primary use
Synthea / SyntheticMass	Synthetic	Open/free synthetic patient records; FHIR/CSV	Immediate	Ingestion + architecture tests
NHANES	Population laboratory survey	Public CDC laboratory data	Immediate	Normal variation + anomaly research
MIMIC-IV v3.1	Hospital/EHR	Credentialed user + CITI + DUA	Early	Clinical modeling + outcomes
eICU-CRD v2.0	Multicenter critical care	Credentialed user + training + DUA	After MIMIC pipeline	External validation
All of Us	Longitudinal EHR/measurements	Institutional DURA + researcher authorization; Registered/Controlled tiers	Advanced	Generalization + broader population
N3C	Clinical network/EHR	Controlled access through NIH environment	Advanced	Multi-site clinical research
TCIA	Medical imaging	Public de-identified imaging collections	Imaging phase	CT/MRI/PET cancer imaging
UK Biobank	Population + imaging	Application/eligibility; current new applications paused as of Jul 2026	Later	Long-term multimodal validation

34. Future Multimodal Expansion Strategy

The laboratory-only stage is a deliberate research constraint, not the final technical limit. The architecture should eventually support multiple evidence modalities through separate encoders and a shared patient-state layer.

34.1 Modalities to add later

- Radiology: CT, X-ray, MRI, PET/other imaging as appropriate for the research question.
- Digital pathology: whole-slide images and pathology-derived features.
- Physiological signals: ECG, pulse oximetry, wearable time series where access permits.
- Genomics/omics: only under appropriate access, consent, governance and research design.
- Clinical text: reports/notes only when access, privacy, and de-identification allow it.

34.2 Multimodal architecture

Use modality-specific encoders -> modality quality/availability assessment -> shared patient representation -> task-specific prediction heads. A patient who has only laboratory data should still receive a laboratory-based assessment. When imaging later becomes available, the system should incorporate it as new evidence and state explicitly how the assessment changed.

35. Federated Learning - Expanded Research Design

Federated learning should be treated as a research track for multi-institution collaboration, not as a synonym for privacy. Federated optimization can keep raw patient data local, but it does not automatically prevent all privacy leakage or security threats. Recent healthcare FL reviews continue to identify heterogeneity, calibration, external validation, privacy safeguards, communication cost and translational readiness as major issues. [22-24]

35.1 Research sequence

1. Simulation: split a public/credentialed dataset into synthetic hospital clients with controlled non-IID distributions.
2. Federated baseline: FedAvg with the same architecture as the centralized benchmark.
3. Heterogeneous availability: clients intentionally observe different laboratory subsets.
4. Personalization: FedProx/personalized layers or other validated approaches.
5. Privacy/security: secure aggregation and differential privacy experiments with utility/privacy trade-off measurement.
6. Real multi-site study: only after governance, agreements, local validation and security controls exist.
7. Adaptive FL: investigate whether sites can share model updates while keeping local test-availability policies private.

35.2 Federated metrics

- Global performance
- Per-site performance
- Worst-site performance
- Cross-site calibration
- Subgroup fairness
- Communication volume
- Rounds to convergence
- Training/inference cost
- Privacy budget where DP is used
- Model drift and site failure robustness

35.3 Recommended Python tooling

Flower is a practical starting point because its documentation is explicitly oriented toward researchers and developers bringing existing ML workloads into a federated setting. [25] FATE is an alternative framework worth evaluating for more enterprise-oriented FL workflows. The research code should keep the model implementation independent from the federation framework so that centralized and federated experiments use the same underlying model definition.

36. Agentic Clinical Intelligence - Safe Research Design

Agentic AI belongs around the validated predictive stack, not inside the clinical truth source. The deterministic/validated ML models should generate predictions; agents should coordinate tools, retrieve evidence, detect missing inputs, route tasks, perform consistency checks, and draft human-readable reports.

36.1 Proposed agents

- Ingestion Agent - accepts documents/data and verifies parsing.
- Evidence Mapping Agent - maps observations to standardized concepts and provenance.
- Quality Agent - checks completeness, units, timestamps and impossible values.
- Analysis Router - chooses validated models appropriate to the available data.

- Uncertainty Agent - collects model uncertainty, data coverage and abstention state.
- Evidence Acquisition Agent - proposes additional evidence based on validated rules/EVI models.
- Research Agent - retrieves approved literature/standards and attaches citations to reports.
- Verification Agent - checks that claims in the generated report are supported by model outputs and approved sources.
- Report Agent - converts structured results into a clinician-readable report.
- Audit Agent - records every tool call, source, model version and user override.

36.2 Hard boundaries

- No autonomous diagnosis claim beyond validated task definitions.
- No autonomous treatment prescription.
- No autonomous laboratory ordering in the research prototype.
- No fabrication of missing measurements or clinical history.
- No unverified medical claim generated solely by an LLM.
- Every recommendation must be traceable to a rule, model output, approved evidence source or explicit human input.

37. Research Statistics and Experimental Analysis Package

37.1 Primary endpoints by track

Track	Primary endpoint family
Anomaly detection	Detection utility, false-alarm burden, stability under normal variation, agreement with defined reference labels when available.
Binary prediction	AUROC, AUPRC, sensitivity, specificity, PPV, NPV, F1, calibration, Brier score.
Time-to-event	Concordance, calibration, time-dependent AUC, Brier score and clinically meaningful time horizons.
Adaptive acquisition	Uncertainty reduction, performance gain per additional test, number of requests, stopping efficiency.
Federated learning	Global + site-specific performance, calibration, worst-site performance, communication and privacy/utility trade-offs.
Multimodal	Incremental value over laboratory-only baseline, subgroup robustness, modality-ablation analysis.

37.2 Statistical safeguards

- Pre-specify the primary endpoint and primary model comparison before final testing.
- Use patient-level splits; never split multiple observations from one patient across train/test in a way that leaks identity.
- Use temporal or site-based holdouts where feasible.
- Use bootstrap or appropriate resampling for confidence intervals.
- Correct or control for multiple comparisons where large model/feature searches create multiplicity risk.
- Perform calibration analysis, not only discrimination analysis.
- Report missingness patterns and subgroup performance.
- Perform sensitivity analyses for cohort definition, index windows, imputation and label noise.

38. Research Reporting, Publication and Supplementary Materials

For each manuscript, maintain a publication package that can be reconstructed from the experiment registry. TRIPOD+AI should guide prediction-model reporting, and PROBAST+AI should be used as a risk-of-bias/applicability self-audit. For diagnostic-accuracy work, use the applicable STARD/STARD-AI guidance; for interventional clinical trials, use the applicable SPIRIT-AI/CONSORT-AI guidance.

38.1 Standard paper package

- Protocol and analysis plan version.
- Dataset/access statement and dataset version.
- Cohort flow diagram.
- Variable/feature table.
- Model specification table.
- Primary and secondary results tables.
- Calibration and discrimination figures.
- Subgroup/fairness table.
- Error-analysis appendix.
- Model card and risk-of-bias checklist.
- Code/reproducibility statement and permitted artifact release plan.

39. Clinical Safety Case and Regulatory Preparation

From the research stage, maintain a safety case explaining intended use, foreseeable misuse, hazards, mitigations, evidence and residual risk. Under EU MDR Rule 11, software intended to provide information used for diagnostic or therapeutic decisions can fall into Class IIa, IIb or III depending on the impact of the decisions supported. The European Commission also published MDCG 2019-11 rev.1 in June 2025 on software qualification and classification. [16,17]

39.1 Safety artifacts to maintain

- Intended-purpose statement for every version.
- Clinical claim matrix: what the model does and does not claim.
- Hazard analysis and FMEA-style risk register.
- Data-quality failure analysis.
- Algorithmic failure modes and abstention policy.
- Human-factors assumptions and usability evidence.
- Cybersecurity threat model.
- Post-deployment monitoring concept for any future clinical deployment.
- Change-control policy for model/data updates.

WHO guidance on AI for health places ethics and governance, human rights and accountability at the center of health AI design. This should be reflected in the project governance package from the first research cycle rather than added at deployment. [28,29]

40. 36-Month Integrated Research Roadmap

Period	Research phase	Primary outputs
Months 1-3	Scientific foundation	Protocol, literature review,

Period	Research phase	Primary outputs
		Synthea/NHANES ingestion, canonical lab schema, research registry, initial anomaly baseline.
Months 4-6	Clinical data foundation	MIMIC access and cohort construction; missingness analysis; baseline supervised models; first manuscript dataset.
Months 7-9	Temporal intelligence	Longitudinal features, GRU/LSTM/Transformer experiments, survival modeling, calibration and external-holdout design.
Months 10-12	External validation	eICU acquisition, locked-model validation, subgroup/fairness analysis, error analysis, manuscript 1/2.
Months 13-15	Adaptive evidence	Abstention, uncertainty, missingness-aware models, masked-test experiments, EVI prototypes.
Months 16-18	Adaptive clinical research	Sequential evidence acquisition simulations, next-test policy evaluation, clinician-readable request/report workflow.
Months 19-21	Federated baseline	Simulated federated hospitals, non-IID labs, variable test availability, FedAvg/FedProx/personalization experiments.
Months 22-24	Agentic orchestration	Bounded agents, evidence retrieval, model routing, verification/audit layer, human-in-the-loop tests.
Months 25-27	Broader population	All of Us or another approved longitudinal resource, transportability and generalization study.
Months 28-30	Imaging preparation	TCIA imaging pipeline, DICOM QA, modality encoder prototypes, laboratory-only vs imaging-only baselines.
Months 31-33	Multimodal research	Lab + imaging fusion, missing-modality experiments, federated multimodal simulation if feasible.
Months 34-36	Clinical research maturity	Silent/prospective validation planning, safety case refinement, final manuscripts, model cards, reproducibility release.

41. Immediate Start Package - First 30 Working Days

1. Create the Git repository and research document tree.
2. Create the research protocol and preregistration draft.
3. Build a literature-search log and evidence matrix.

4. Register Synthea and NHANES data sources and record versions/access dates.
5. Implement canonical lab schema and metadata-driven analyte registry.
6. Implement unit normalization and reference-range handling.
7. Implement observation-status and missingness-mask representation.
8. Build the first CBC anomaly benchmark: robust z-score + Isolation Forest + autoencoder baseline.
9. Create automated evaluation and report generation.
10. Start PhysioNet credentialing/CITI/DUA workflow for MIMIC-IV and eICU.
11. Create the experiment registry and model-card templates.
12. Create the adaptive-evidence simulation skeleton: hide selected measurements -> predict -> score uncertainty -> request a next evidence candidate -> reveal -> update.

42. Research Decision Gates

Do not advance to the next technical layer merely because a model is more complex. Advance only when a research gate is met.

Gate	Decision area	Minimum evidence
Gate 1	Data integrity	Schema, provenance, units, missingness and cohort checks pass reproducibly.
Gate 2	Baseline validity	Simple baseline models establish a trustworthy benchmark.
Gate 3	Clinical relevance	The target label/outcome has a defensible clinical definition and index window.
Gate 4	Generalization	Performance is stable on a genuinely held-out cohort/site/time period.
Gate 5	Uncertainty	The model can identify low-confidence cases and calibrate probabilities.
Gate 6	Adaptive value	Additional-test recommendation improves information or performance enough to justify the complexity.
Gate 7	Agent safety	Agent outputs are traceable and verified against structured evidence.
Gate 8	Federated validity	FL provides useful cross-site behavior under realistic heterogeneity and governance.
Gate 9	Multimodal readiness	The laboratory-only branch is stable enough to serve as a comparator when imaging is introduced.

43. Updated Research Architecture

The final target is a modular clinical intelligence platform in which evidence availability is dynamic. The laboratory branch remains the foundation, while the same patient-state layer can later accept imaging and other validated modalities.

1. Evidence ingestion: PDF, CSV, FHIR, structured lab feed, later DICOM and other modalities.
2. Evidence normalization: concepts, units, dates, reference context and provenance.
3. Evidence coverage engine: observed, missing, pending, stale, invalid, not applicable.
4. Patient state engine: current state + historical trajectory + derived phenotype representation.
5. Model layer: anomaly, classification, survival, temporal, multimodal and calibrated ensemble models.
6. Uncertainty layer: confidence, calibration, abstention and data-sufficiency estimation.
7. Adaptive acquisition layer: evidence-gap detection + EVI/next-test prioritization.
8. Agentic orchestration layer: tool use, literature retrieval, verification, reporting and audit.
9. Federated layer: optional distributed training and cross-site validation.
10. Human interface: clinician dashboard, trend views, evidence request queue, explanation and audit trail.

44. Current Source and Access Notes - Verified September 5, 2026

PhysioNet - MIMIC-IV v3.1. [Source](#) - Credentialed access requires credentialed-user status, CITI Data or Specimens Only Research training, and a data use agreement.

PhysioNet - eICU-CRD v2.0. [Source](#) - Credentialed access; required human-research training and DUA. Useful for external/site validation.

CDC/NCHS - NHANES. [Source](#) - Public continuous-survey data with laboratory components; use for population laboratory analysis and reference variation.

Synthea. [Source](#) - Synthetic patient histories available in FHIR/CSV and other formats; suitable for architecture and pipeline development.

All of Us - Data Access Tiers. [Source](#) - Public, Registered and Controlled tiers; Registered/Controlled require researcher authorization and institutional DURA.

All of Us - Workbench. [Source](#) - Current workbench documentation; current CDR v9 collections are listed in the support documentation.

UK Biobank - Access. [Source](#) - As of July 1, 2026, new applications were paused; UK Biobank stated an intention to accept new applications in late 2026.

TCIA - De-identification. [Source](#) - Public imaging collections are de-identified using standards-based DICOM/TCIA processes.

EU MDR 2017/745. [Source](#) - Rule 11 addresses software providing information used in diagnostic/therapeutic decisions and sets risk-based classes.

MDCG 2019-11 rev.1. [Source](#) - European Commission page for the June 2025 software qualification/classification guidance.

WHO AI for Health Ethics and Governance. [Source](#) - Ethics, governance, accountability, human rights and safety guidance for AI in health.

TRIPOD+AI. [Source](#) - Current reporting guidance for clinical prediction models using regression or AI/ML methods.

PROBAST+AI. [Source](#) - Tool for quality, risk of bias and applicability assessment of prediction models using regression or AI methods.

Flower. [Source](#) - Federated-learning framework and tutorials for researchers/developers.

Federated learning methodological review. [Source](#) - Reviews methodological challenges including non-IID data, missingness, distribution shift and deployment concerns.

Federated learning systematic review. [Source](#) - Large systematic review of healthcare FL applications; reports that real-world clinical deployment remains limited.

2026 FL methodological gaps review. [Source](#) - 2026 review emphasizing heterogeneity, privacy, evaluation and governance gaps.

45. Final Research Position

The defining capability of MIR Medical Intelligence should be evidence-adaptive reasoning rather than a fixed disease checklist. A model receives whatever the current record contains, analyzes it immediately, identifies abnormalities and trajectories, estimates risk only for tasks supported by sufficient evidence, communicates uncertainty, and identifies the next evidence that could materially improve the assessment. The system should expand organically from common laboratory tests to specialized assays, then to longitudinal EHR and potentially genomics/omics, and ultimately to imaging and multimodal reasoning.

The core scientific contribution should therefore be evaluated at three levels: (1) whether laboratory and longitudinal patterns contain useful predictive signal; (2) whether uncertainty-aware, missingness-aware and adaptive evidence acquisition improves reliability; and (3) whether these techniques remain useful across institutions, modalities and federated environments. This creates a research program that can grow for years without locking the platform to a static panel of tests or a single model family.